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CHAPTER 12

RECIPES FOR HORROR IN GRAECO-ROMAN MAGIC AND MEDICINE

Sean Coughlin

A certain atmosphere of breathless and unexplainable dread of outer, unknown forces must be present; and there must be a hint, expressed with a seriousness and portentousness becoming its subject, of that most terrible conception of the human brain – a malign and particular suspension or defeat of those fixed laws of Nature which are our only safeguard against the assaults of chaos and the daemons of unplumbed space.

H. P. Lovecraft, 'Supernatural Horror in Literature', *The Recluse*, 1927: 24

Introduction¹

I am not sure whether any writers in the ancient Mediterranean believed there were 'fixed laws of Nature' as Lovecraft understands them. For many ancient Greek and Roman writers, even philosophers and the doctors historians call 'rational', the natural world was governed by intentional or quasi-intentional forces acting according to their own non-moral impulses. 'The world is full of gods', as Aristotle says.² Still, there were practices – call them magic, sorcery, witchcraft, or even medicine – that aimed at persuading daemons or similar forces to carry out the wishes of the person invoking them. If we think of horror as a fear that the paranormal can slip into an otherwise predictable world, then practices like these can be understood as ways of coping with or even exploiting those forces.³

The recipes of my chapter's title are those that presume the 'daemons of unplumbed space' and record practical steps and ingredients to protect against them or conjure them up. The steps may involve mixing elixirs, potions and philtres, casting spells, and reciting incantations. The ingredients range from rare and expensive varieties of incense to the disgusting and abject materials of *Dreckapotheke* or filth pharmacy, including blood, urine, faeces, sexual fluids, and parts of animal or human corpses. Such ingredients are commonplaces of supernatural horror today and they are commonplaces of magical and medical writing from Graeco-Roman antiquity. I could only guess at the precise lines of transmission between ancient texts and modern horror literature.⁴ The Greek and Egyptian magical papyri studied today were mostly buried until the nineteenth century; and even after they emerged, they remained difficult to access as ancient Egyptian writing systems were still in the process of being deciphered.⁵ Still, it seems plausible to me that the formulae they contain are ancestors of the kinds we find described in supernatural horror, from *Macbeth's* weird sisters' eye of newt to the tannis root charm of *Rosemary's Baby*.

The magical and medical writings of the Graeco-Roman period offer glimpses of how people understood the worlds of magic and the everyday to be related, and some glimpses come not through abstracted magical practice itself, but through stories of different actors' tangible encounters with magic recipes.⁶

In this chapter, I want to explore some of these stories and some of these encounters, casting aside a broader exploration of magic's spiritual and allegorical dimensions to concentrate on the bodily manifestations and experiences elicited by ancient recipes for horror.⁷ My focus is on the human and animal body, not merely as something subject to horror, but as a medium and even active participant in the rituals prescribed. The investigation will be framed by two encounters: first, Emperor Hadrian's witnessed exhibition (*epideixis*) of a magical procedure by Pachrates, the high priest of Heliopolis, offering a glimpse into the ritualistic and performative aspects of magical practice; and second, my own encounter with replicating a recipe for myrrh-blood ink, providing a modern perspective on the chemical and sensory dimensions of ancient magic. The aim of the replication is not only to 'experience the past' for its own sake, but serves two additional purposes. First, as scholars are becoming increasingly aware, familiarizing oneself with the material reality behind the text of the recipe provides additional evidence for understanding how it is supposed to work (whether in fact it works or not) and consequently how best to interpret what the text says.⁸ Second, as William Tullett has recently argued in his work on sensory studies, by replicating or recreating things similar to what was experienced in the past, we are able to note the ways our experiences and associations differ from those recorded in our sources, and then use the question 'why do these differences exist?' as a starting point for historical reflection.⁹ Both approaches offer novel ways of studying recipes for horror not merely as texts but also as records of complex interactions involving the practitioner, ingredients, procedures and victims. Through these encounters, we can discern distinct layers of meaning, fears, and fascination that these recipes embody.

Categorizing recipes

Before moving to the encounters, it will be helpful to have an overview of the categories ancient authors used to order magical practices and recipes, and the easiest way to do this is to begin with an example of the kind of texts we are dealing with:

Agathokles' dream-sending procedure: Take a completely black cat that died a violent death, fashion a strip of papyrus and inscribe on it with myrrh the following things and the [name of the] person to whom you want to send a dream, and place it into the mouth of the cat.

ὄνειροπομ[πὸ]ς Ἀγαθοκλεὺς. λαβὼν αἴλουρον ὀλομέλανα βιοθ[άν]ατον, π[ο]ιήσας
πιττάκιον καὶ ἐνγράψας ζζ τὰ ὑποκείμενα καὶ ὃν θέλεις ὄν[ειροπ]ομπέισαι καὶ
ἐνθεῖς εἰς τὸ στόμα τοῦ αἰλούρου.

GEMF 15 156–8 = PGM XII 107–9 = 84,1–3 FTT,
tr. Dieleman, Sarischouli et al.

The rite described here would not seem out of place in a tale of supernatural horror. The dream-sending procedure is attributed to a mysterious and otherwise-unknown Agathokles, whose attribution gives the procedure authority and credibility. Many of the elements in his procedure – the black cat, a violent death, magic ink, manipulation of corpses – are common in the genre of supernatural horror. A series of *voces magicae* or magic words, another familiar element, immediately follows the quotation. These are meant to invoke gods and demons, who are then commanded to send whatever information one wants to whatever addressee one wishes through a dream.

The papyrus transmitting this recipe, Leiden I 384, first came to the attention of European scholars in 1828; given its recent reception, therefore, it is remarkable how familiar its elements seem.¹⁰ Of course, whenever an ancient text seems familiar, we should keep in mind that it may be because we are reading our own modern expectations into the text. There are however *prima facie* reasons for thinking that the recipes share a family resemblance with the genre of supernatural horror, for there is something akin to Lovecraft's dread in many of the recipes we find in the Greek and Egyptian formularies. For example, written on the same papyrus, we find the following incantation:

Come to me, lord of forms, and awaken men and women for me; compel them by your everstrong and mighty power to do all the things that are both written and spoken by me. EISAPHSANTA PHOUREI ARNAI. . . . SUNPHREŌ RIŌBAI OSOI, you are (?) ATEPHTHO AŌREL ADŌNAI. Fill them with terror and trembling, flutter their minds with fear of you, and do to him, NN, all the things that have been prescribed.¹¹

ἤκαί μοι ὁ δεσπότης τῶν μορφῶν καὶ δειέγειρόν μοι ἄνδρας καὶ γυναῖκας, ἀνάγκασον αὐτοὺς ποιῆσαι τῇ σαι εἰσχυρᾷ καὶ κρατα δυνάμει πάντα τὰ ὑπ' ἐμοῦ γραφόμενά τε καὶ λεγόμενα. εἰσαψαντα φουρει αρναι: . συνφρεω ριωβαι ρσοι συ ατεφθο αφρελ Αδωναι. καὶ ποιήσον αὐτοὺς ἐντρόβους, ἰντρόμους, ἐπτοημένους τὰς φρένας ποιήσας διὰ τὸν φόβον σου καὶ ποίει τῷ Δ' ἅπαντα τὰ προγεγραμμένα.

GEMF 15 99–104 = PGM XII 51–5 = 76,11–16 FTT,
tr. Dieleman, Sarischouli et al.

The incantation, addressed to the god Eros, is part of a larger procedure for invoking the god as an attendant (*paredros*, πάρεδρος), an entity at least superficially similar to a familiar spirit.¹² The more important similarity, however, is how the scene is described. The attendant is invoked in order to coerce a victim to do something. The means of coercion are terror and fear. The attendant brings about *enphobos* (ἐνφοβος), a state of fear (the term is only found in magical texts); *entromos* (ἴν-, ἔντρομος), a state of trembling; and *epitoēmenos tēn phrena* (ἐπτοημένος τὴν φρένα), a fluttering or terrifying of the mind *dia ton phobon sou* (διὰ τὸν φόβον σου), 'through fear of you', i.e. the god.¹³ To these, one could add similar states and emotions, like *phrikē* (φρίκη), the shuddering of the body, mentioned in other procedures.¹⁴ The supernatural attendant, therefore, is portrayed in the procedure as an object and cause of fear and dread.

There are many kinds of procedures that involve the appeal to such forces. There are general terms for them, such as spells (*epilogoi*, ἐπίλογοι), formulae (*logoi*, λόγοι), incantations (*epōidai*, ἐπωδαί), rites (*teletē*, τελεταί) and potions (*posima*, *potēria*; πόσιμα, ποτήρια); and there are specific terms for specific procedures: bindings (*katadesmoi*, κατάδεσμοι), attraction producers (*agōgai*, *agōgima*; ἀγωγαί, ἀγωγήμα), dream-senders (*oneiropompoi*, ὄνειροπομποί), sleeplessness-producers (*agrupnētika*, ἀγρυπνητικά), invisibility procedures (*amaurōseis*, ἀμαυρώσεις), love charms (*philtrā*; φίλτρα), love-bindings (*philtrokatadesmoi*, φίλτροκατάδεσμοι), *iunx* or *jinx* (ἰνυξ) for attraction,¹⁵ inflaming procedures (*empura*, ἔμπυρα), charisma procedures (*charitēsia*, χαριτήσια), procedures to produce discord between people (*diakopoi*, διακοποί), procedures for restraining someone's anger (*thumokatokha*, θυμοκάτοχα), spells using slander against a divinity or demon (*diabolai*, διαβολαί), and phylacteries (*phylaktēria*, φυλακτήρια), procedures for protection from others' magic or from gods and demons. This list is only a selection but, as we will see, it corresponds well with what we find in literary and documentary sources contemporary to the magical papyri, and will be helpful for understanding what follows.

Encounter I: The Imperial Audience at Heliopolis (GEMF 57 = PGM IV 2441–620)

The Recipe

Attraction procedure (*agōgē*). Equipment for a lunar burnt offering. It attracts those who are uncontrollable and without the use of magical material on the same day; it inflicts sickness excellently and destroys powerfully, sends dreams beautifully, accomplishes dream revelations marvellously and in its many demonstrations has been marvelled at for having no failure in these matters.

Burnt Offering. Exhibited by Pachrates, the high priest of Heliopolis, who exhibited the power of his divine magic to the Emperor Hadrian. It attracted in one hour, made sick in two hours, destroyed in seven hours, and it sent dreams to the emperor himself when he was testing the complete truth of his divine magic. He was so amazed at the high priest, that he ordered twice the pay to be given to him.

Ἀγωγή· σκευὴ ἐπιθύματος σεληνιακοῦ ἄγουσα ἀσχέτους καὶ ἀνουσιάστους μονοήμερους, κατακλίνει γενναίως καὶ ἀναιρεῖ ἰσχυρῶς, ὄνειροπομπεῖ καλλίστως, ὄνειραιτητεῖ θαυμαστῶς καὶ ἐν πλείστοις ἀποδείξεσιν ἐθαυμάσθη οὐδεμίαν ἔγκλισιν ἔχουσα τούτων.

ἐπιθύμα· ἐπεδείξατο Παχράτης, ὁ προφήτης Ἡλιουπόλεως, Ἀδριανῷ βασιλεῖ ἐπιδεικνύμενος τὴν δύναμιν τῆς θείας αὐτοῦ μαγείας. ἤξεν γὰρ μονώρον, κατέκλινεν ἐν ὥραις β̄, ἀνείλεν ἐν ὥραις ζ̄, ὄνειροπόμπησεν δὲ αὐτὸν βασιλέα ἐκδοκιμάζοντος αὐτοῦ τὴν ὅλην ἀλήθειαν τῆς περὶ αὐτὸν μαγείας· καὶ θαυμάσας τὸν προφήτην διπλὰ ὀψώνια αὐτῷ ἐκέλευσεν δίδοσθαι.

GEMF 57 = PGM IV 2441–55 = 82–3 Betz, tr. O'Neil, modified

While nearly any of the recipes in the *Greek and Egyptian Magical Formularies* would be at home in a paper on horror, this procedure from *GEMF* 57 ('The Paris Magical Codex') is interesting in terms of its complexity, its parallels in other procedures, and its performative aspect.¹⁶ It is an attraction procedure (*agōgē*, ἀγωγή) that combines a burnt incense offering (*epithuma*, ἐπίθυμα), followed by a slander spell (*diabolē*, διαβολή) addressing the lunar goddess Selene,¹⁷ then by a protection spell or phylactery (*phulaktērion*, φυλακτήριον), and finally some further incantations to Selene. It is one of two attraction procedures in the Paris codex that involve a slander spell addressing Selene (or Selene-Hecate-Artemis-Aphrodite). The other follows immediately after (at *GEMF* 57 = *PGM* IV 2618–705), with spells to Selene that differ only slightly in phrasing and a phylactery and incense offering that differs more noticeably.

The text begins with a statement of what the procedure is for, but it does not say who the victim is. In the opening statement, it says the procedure attracts a victim, inflicts illness, and kills (along with some dream sending and prophecy). Only later do we learn a woman is meant, when the victim is introduced as 'she so-and-so'.¹⁸ She is brought to the practitioner when he informs the goddess that this woman has slandered her and desecrated her mysteries, and then suggests to the goddess to send her to him so he can punish her.

A staging of erotic horror

I begin with the opening lines of the procedure as well as one of its incantations as an instance of a recipe that is a cause of horror. I will then go on to discuss its phylactery as a cause of horror prevention, and then to its burnt offering of filthy materials as an instance of horror itself.

The procedure is unique in beginning with a story of its use. The story describes an occasion when the procedure was used in the context of an *epideixis* (ἐπίδειξις), a word that means 'exhibition' or 'display', the kind of thing one does to make something manifest to others. Such a story recalls a long tradition going back at least to the middle Kingdom (beginning in the 2000s BCE) of sorcerers or priests being asked by the king or pharaoh to prove their skill through a public display.¹⁹ One example comes from the Westcar papyrus (P. Berlin 3033, c. 1600s BCE), where Djedi, a commoner and magician, is asked by king Khufu (Cheops) to display his ability to tame wild animals and reattach heads that have been severed from their bodies.²⁰ In a second-century CE Greek-speaking context, however, *epideixis* also has a technical sense, referring to public speeches associated with the orators of the Second Sophistic. The sophists of this period are known for composing and performing set-speeches to display their rhetorical virtuosity to audiences that would include respected members of society. We can see a parallel in Pachrates' case, whose *epideixis* was for the Emperor Hadrian. Like in the early Egyptian tradition, however, what is on display is not a rhetorical speech, but the magic of a practitioner. Since magic is not something that can be made apparent in itself, it must be made manifest through a medium or victim, in this case the woman who serves as proof of the magic's power. *Epideixis*, therefore, here implies both the public spectacle and the effects of magic on the body of the victim as is its main event.

According to the author, the *epideixis* was conducted by Pachrates, the *prophetēs* or high priest of Heliopolis.²¹ It seems likely that no such event took place. Still, it is worth imagining it did in order to consider the reasons why someone might have written a story as if it did. Like the dream-sender of Agathokles discussed earlier, this attraction procedure, specifically the *epithuma* or incense offering, is attributed to an expert and authority.²² One would rightly be tempted to think this is done to give the recipe credibility, but this is only part of what is going on. Unlike the reference to Agathokles in the previous procedure, in this case, what seems most important is not the authority of the practitioner, but the fact that the procedure was proven by a thorough test (*ekdokimeion*, ἐκδοκιμεῖον). The credibility of the procedure therefore does not derive solely from Pachrates' authority, but primarily from its being tested and proven.

There are at least two ways we might understand the relation of Pachrates' *epideixis* to the kind of testing prompted by Hadrian's audience. One way, recently suggested by Radcliffe Edmonds III, is to consider that Hadrian's 'test' consists in Pachrates teaching him how to perform the procedure and the story is recounting Hadrian's own use of the magic spells. On this view, it is the success of Pachrates' magic in the hands of someone from 'outside the temple gates' that is a sign of its power.²³ Hadrian, meanwhile, would be being portrayed as another civic leader with an interest in violent magic, like Cleopatra VII, who is associated with *iunx* procedures, or Attalus III, king of Pergamum (ruled 138–133 BCE) and Mithridates VI Eupator, king of Pontus (ruled 123–63 BCE), who are reported to have tested magic and poisons on criminals sentenced to death.²⁴ Yet, for reasons that will become clear, it seems unlikely to me that this text suggests Hadrian carried out any procedures himself.

A second possibility is to consider the relation between Pachrates *epideixis* and the emperor's request for proof along the lines of the earlier Egyptian tradition, where a magician exhibits their expertise to the civic leader. Support for this reading comes from the text itself: Pachrates is said to have sent dreams to the emperor, rather than the emperor sending them to himself, and so we can suppose that Pachrates carried out the other procedures as well.

More telling evidence, however, comes from parallels in contemporary sources where *epideixis* as public performance is portrayed as a means of giving proof and establishing reputation. One such source is Galen of Pergamum, a rationalist doctor and medical writer who lived in the second half of the second century CE and so is contemporary with Pachrates and Hadrian. Galen is a notoriously shameless boaster and polemicist. While this often makes him frustrating to read, it also means we have reports from his own hand of the medical and anatomical procedures that he performed in public. The way he refers to these performances is as *epideixeis*. Claire Bubb, Luis Salas and Heinrich von Staden have argued that Galen's use of *epideixis* and related terms suggests he is thinking of his performances as similar theatre or a public speech.²⁵ As von Staden puts it, Galen 'evocatively inscribes his public anatomical exhibitions in a tradition of theatrical rhetorical virtuosity with which his audiences were thoroughly familiar'.²⁶ So, for example, in his work *On My Own Books*, he talks about how his friends compelled him to return to performing dissections of animals for public audiences. The reason, he says,

is so that he could show (*epideixōn*, ἐπιδείξων) through the dissection that he had not falsified anything (*emauton ouden epseusmenon*, ἐμαυτὸν οὐδὲν ἐψευσμένον) in his anatomical writings – a performance he says that went on for days.²⁷ He would ask his audience (often from among the city's intellectuals and politicians) to pick an anatomical detail they would like to see from a competing anatomist's book. For each choice, he would show the existence of the structures he wrote about and their functions. He would dissect the animal body (goats or pigs, often alive), show the structure, and in doing so establish the superiority and truth of his own anatomical writings.²⁸ *Epideixis*, therefore, has a meaning beyond the familiar one of performing a set speech at this period. It includes a public 'showing' of the truth of one's knowledge to settle the status of one's expertise. Since the knowledge is about anatomy, a written treatise is not sufficient to give it credibility. Like in Pachrates' case, it also needs to be manifested through the body of the victim.

The second contemporary source is Lucian. The last story in Lucian's *Philopseudes* is the inspiration for Goethe's *Der Zauberlehrling* and Disney's *The Sorcerer's Apprentice*.²⁹ It is about an Egyptian holy man (*anēr hieros*, ἀνὴρ ἱερός) in Memphis: clean shaven, imperfect Greek (*ou katharōs hellēnizonta*, οὐ καθαρῶς ἐλληνίζοντα), but wonderfully wise with a complete Egyptian education (*thaumasios tēn sophian kai tēn paideian eidōs tēn Aigyp̄tion*, θαυμάσιος τὴν σοφίαν καὶ τὴν παιδείαν πᾶσαν εἰδὼς τὴν Αἰγύπτιον).³⁰ His name is Pancrates (Παγκράτης), which means, 'All Powerful', but which some scholars point out is only a slight variation in the Hellenization of 'Pachrates' from the Paris codex.³¹ In Lucian's story, the characters tell how they became Pancrates' students after they saw him perform (*ergazesthai*, ἐργάζεσθαι) many marvellous things (*polla terastia*, πολλὰ τεράστια), including riding on crocodiles, swimming with other animals (like Djedi in the story of Khufu from the Westcar papyrus), and turning brooms, pestles and other things around the house into servants and attendants.³² While Lucian does not explicitly call the performance an *epideixis*, it is still something witnessed by the public, and it is its marvellous character that leads the story's protagonist to become his student. The procedure, however, is not something foolproof that can be done even by someone from 'outside the temple walls.' Famously, the point of the story is his student's inability: he cannot turn off the spell once he has turned the pestles (or broomsticks in Disney's version) into servants.

Therefore, parallels in both Galen and Lucian suggest a context where *epideixis* are public displays. In Galen, such displays are means of proving the practitioner's skill, while in Lucian, even if not called *epideixis* directly, public performance is a means of establishing authority. Likewise with Pachrates' burnt offering, the public exhibition is meant to prove the procedure works.

While it seems more likely to me that the text implies the proof was performed by Pachrates, even if it were Hadrian, the dream sending is only one part of the procedure. The main part involves acts against the body of the victim: attracting her, causing her illness, and causing her death. We are even given specific details of this. On this occasion, it attracted the victim in one hour, made her sick in two, and destroyed her in seven. The details may seem artificial. An increase in the time of the procedure corresponding to an

increase in the intensity (from attraction to death) is paralleled in, for example, a Demotic procedure in *GEMF* 16 for a similar effect. This recipe says one must stand over the severed head of a donkey, anoint one's hands and mouth with donkey blood, and recite spells for four days to make the victim sick, seven to make them die.³³ Pachrates' procedure is conveniently quicker, which helps to make the performative aspect more plausible. Yet, regardless of how common the idea is of a proportion of time to intensity, the procedure manifests in the body of the victim. There is no mention that any supernatural being appears or indeed that there is any other manifest proof that the procedure works. There is Pachrates, with his special recipe for a burnt offering, and there is the victim's body, as proof that it is effective.

The centrality of the body is implied by the hymn to Selene (here also called Aktiōphis, origin unknown), which describes the effect that Pachrates' audience would have witnessed:

Spell: 'Let all the darkness of clouds be dispersed for me, and let the goddess Aktiōphis shine for me, and let her hear my holy voice. For I come announcing the slander of NN, a defiled and unholy woman, for she has slandered your holy mysteries to the knowledge of people. She, NN, is the one, [not] I, who says, "I have seen the greatest goddess, after leaving the heavenly vault, on earth without sandals, sword in hand, and [speaking] a foul name." It is she, NN, who said, "I saw [the goddess] drinking blood." She, NN, said it, not I, AKTIŌPHIS ERESCHIGAL NEBOUTOSOUALĒTH PHORPHORBA SATRAPAMMŌN CHOIRIXIĒ, FLESHEATER. Go to her NN and take away her sleep and put a burning heat in her soul, torment and stinging in her heart, and banish her from every place and from every house, and attract her here to me, NN.' ... After saying these things, sacrifice. Then raise loud groans and go backward as you descend. And she will at come at once. But pay attention to the one being attracted, so that you may open the door for her; otherwise the spell will fail.

λόγος· «διασταλήτω μοι πᾶσα νεφῶν σκοτία, καὶ ἐπιλαμψάτω μοι ἡ θεὸς Ἀκτιῶφίς καὶ ἀκουσάτω μου τῆς ἱερᾶς φωνῆς·ωνῆίνω γὰρ καταγγέλλων τὴν διαβολὴν τῆς μιαρᾶς καὶ ἀνοσίας, τῆς δεῖνα· διέβαλεν γάρ σου τὰ ἱερὰ μυστήρια ἀνθρώποις εἰς γνῶσιν. ἡ δεῖνά ἐστιν ἡ εἰποῦσα ὅτι – <οὐκ> ἐγὼ εἰμι εἰποῦσα ὅτι· ἐγὼ ἶδον τὴν μεγίστην θεὸν καταλιποῦσαν τὸν πόλον τὸν οὐράνιον, ἐπὶ γῆς γυμνοσάνδαλον, ξιφηφόρον, ἄτοπον ὄνομα <όνομά>σασαν· ἡ δεῖνά ἐστιν ἡ εἰποῦσα· ἐγὼ τὴν θεὸν αἶμα πίνουσιν· ἡ δεῖνα εἶπεν, οὐκ ἐγὼ, ΑΚΤΙΩΦΙ ΕΡΕΣΧΙΓΑΛ ΝΕΒΟΥΤΟΣΟΥΑΛΗΘ· ΦΟΡΦΟΡΒΑ· ΣΑΤΡΑΠΑΜΜΩΝ. ΧΟΙΡΙΞΙΗ· ΣΑΡΚΟΒΟΡΑ· βάδισον πρὸς τὴν δεῖνα καὶ βάσταξον αὐτῆς τὸν ὕπνον καὶ δὸς αὐτῇ καὺσιν ψυχῆς, κόλασιν φρενῶν καὶ παροίστηρσιν, καὶ ἐκδιώξασα αὐτὴν ἀπὸ παντὸς τόπου καὶ πάσης οἰκίας ἄξον αὐτὴν ὧδε, πρὸς ἐμέ, τὸν δεῖνα.» ... ἐπὶ δὲ ἀναιρέσεως λέγει· «ἀνάσπασον αὐτῆς τὸ πνεῦμα, κυρία, τῶν μυκτηρῶν τῆς δεῖνα.»

GEMF 57 = PGM IV 2471–99 = 83 Betz, tr. O'Neil, modified

The spell begins by addressing the Aktiōphis/Selene and then informing her that she has been slandered by the victim through the desecration of her mysteries.³⁴ Another series of epithets follows, and then the deity is asked to afflict the victim and lead her to the performer, presumably to be punished. What follows are familiar elements in erotic spells of attraction (*agōgai*).³⁵ The victim is coerced from her house through discomfort, confusion, and erotic madness; her sleep is taken away through an insomnia procedure (*agrypnētikon*, ἀγρυπνητικόν); a burning is caused in her soul (*kausin psychēs*, καῦσιν ψυχῆς) with something like an inflaming procedure (*empyron*, ἐμπυρον); an incense offering is made, which Faraone suggests could be an example of sympathetic causation, the burning incense causing a sympathetic burning in the soul, the torture of which will cause her to leave her house.³⁶ Imagery of erotic torture continues with the ‘torment and stinging of the heart’ (*kolasin phrenōn kai paroistrēsīn*, κόλασιν φρενῶν καὶ παροίστησιν). ‘Stinging’ here translates *paroistrēsis* (παροίστησις), which is both connected to what the gadfly (*oistros*, οἷστρος) does, but is also a general term for frenzy, especially mating-related, hence ‘oestrus’, ‘to be in heat’.³⁷ It is, therefore, an erotic kind of pain that is being afflicted. The erotic torture of the victim is followed by a spell for causing sickness, and finally one that kills her by taking away her breath (*pneuma*, πνεῦμα).

Erotic torture and murder follow the recitation of the slander spell and we should expect that this series of events is part of Pachrates’ *epideixis* to the emperor. The victim’s body takes the place of the goat or pig in Galen’s anatomical dissections, but she is even more the centre of the performance: in Galen’s case, the anatomist himself is at work, while in Pachrates’ exhibition, the agent, the deity, is invisible. The victim’s torment is also primarily bodily. While her soul and heart are afflicted, there seems to be no dualistic assumption of material body and immaterial soul. As Faraone points out, the process of even psychic torture is vividly described in terms of bodily discomforts: sleeplessness, burning, tormenting, stinging.³⁸ A further bodily element is the physical act of performing the ritual, which is the sympathetic act required for the supernatural to bring about its bodily effects. The ritual and its offering are necessary intermediaries between the unseen forces and the effects on the victim, where the operations, especially the burning of incense, mirror the victim’s psychic torment. Even such a condensed passage, therefore, tells us that emperor Hadrian’s encounter with Pachrates’ magic would have consisted in the physical expression of communion with the supernatural and the bodily expression of psychic torment.

Guarding the body from demonic horrors

The next part of Pachrates’ procedure consists in a phylactery, or rite of protection against harm – in this case from Selene, the goddess being invoked:

[The rite] also possesses a protective charm against your falling, for the goddess is accustomed to make airborne those who perform this rite unprotected by a charm and to hurl them from aloft down to the ground. So consequently I have also

thought it necessary to take the precaution of a protective charm so that you may perform the rite without hesitation. Keep it secret. Take a hieratic papyrus roll and wear it around your right arm with which you make the offering. And these are the things written on it: ‘MOULATHI CHERNOUTHAMARO MOULIANDRON, guard me from every evil daimon, whether an evil male or female.’ Keep it secret, son.

ἔχει δὲ φυλακτήριον πρὸς τὸ μὴ σε καταπεσεῖν· σείωθεν γὰρ ἡ θεὸς τοὺς ἀφυλακτῆριαστοὺς τοῦτο πράσσοντας ἀεροφ<ερ>εῖς ποιεῖν καὶ ἀπὸ τοῦ ὕψους ἐπὶ τὴν γῆν ῥίψαι. διὸ οὖν ἀναγκαῖον ἡγησάμην καὶ τοῦ φυλακτηρίου τὴν πρόνοιαν ποιήσασθαι, ὅπως ἀδιστάκτως πράσσης. κρύβε. λαβὼν ἱερατικὸν κόλλημα φόρει περὶ τὸν δεξιὸν βραχίονά σου, ἐν ᾧ ἐπιθύσεις. ἔστι δὲ τὰ γραφόμενα ταῦτα· ΜΟΥΛΑΘΙ· ΧΕΡΝΟΥΘ· ΑΜΑΡΩ· ΜΟΥΛΙΑΝΔΡΟΝ· διαφύλαξόν με ἀπὸ πονηροῦ παντὸς δαίμονος, ἧτοι ἀρσενικοῦ πονηροῦ ἢ θηλυκοῦ· κρύβε, υἱέ.

GEMF 57 = PGM IV 2505–19 = Betz 84, tr. O’Neil, *modified*

The victim’s body is not the only place in which supernatural horror manifests; the practitioner’s body, too, is a potential site. In the case of our spell, the procedure needs to be carried out on a roof, perhaps not the safest place, as the author warns us that the goddess is known to throw people into the air if she is unhappy. One needs, therefore, a recipe for protection.

As with the attraction procedure, the phylactery or protection procedure that accompanies it assumes the daemons of unplumbed space can penetrate the normal workings of the world. It likewise uses a physical act to influence those forces. However, the line between ‘supernatural’ and ‘natural’ (or ‘spiritual’ and ‘bodily’) becomes blurrier than in the case of the erotic attraction spell, as the danger threatened by the spiritual Aktiōphis/Selene is unquestionably bodily. With the attraction procedure, the torture was of the soul and heart, even if expressed in bodily terms, while the illness and death could be conceived of as in some sense spiritual. At least, there is no brutal violence to the body suggested. The phylactery, by contrast, prevents the body from being tossed through the air and broken on the ground. The corporeality of the violence here is much more pronounced.

The knowledge is likewise made to be more embodied. The text emphasizes that this knowledge must be kept secret. Nevertheless, for the knowledge to be effective, the magic words or names mentioned by the procedure cannot merely be uttered. They must be inscribed on the appropriate support. The precise support may vary from procedure to procedure. In this case, the magic words are inscribed on a piece of sacred papyrus, but phylacteries could be made from many materials, such as stone amulets or metal lamella.³⁹ Whatever the symbolic or ritualistic meaning, they are tangible artefacts, which, like the danger itself, suggests an embodiment of hidden magical power.

A further aspect of the phylactery’s embodiment comes from the fact that they must be worn to provide protection. Scholars of Greek, Egyptian and Roman magic note two ways such phylacteries are thought to work, by similarity and sympathy of difference and antipathy. The phylactery provides protection by marking the body of the wearer with

certain sacred and secret qualities. In sympathetic cases, these are qualities like those of the divine entity being called upon, in which protection is afforded through the divine's sympathy with the practitioner. In antipathetic cases, the qualities are opposite or hostile to the divine entity, and so able to frighten it into subjugation.⁴⁰ In either case, the phylactery is a medium through which the practitioner's privileged access to secret knowledge is made corporeal and hence manifest to the divine.

'Fat of a spotted virgin goat, faeces of a dog-faced baboon': Horrific recipes

The final section of the procedure I want to look at is the *epithuma* or incense offering of Pachrates itself. It presents a further element of bodily horror, a horror in itself, and it is helpful to survey some of the ways such recipes and their materials are reported to have been encountered.

Take a field-mouse that's been made divine with spring water; and take two moon beetles made divine with river water, a river crab, the fat of a spotted virgin goat, faeces of a dog-faced baboon, two ibis' eggs, two drachms of styrax, two drachms of myrrh, two drachms of saffron, four drachms of Italian cyperus, four drachms of uncut frankincense, a single onion.

Throw all these into a mortar with the field-mouse and the rest, and having ground them up well, keep them ready for use by storing them in a lead box. And whenever you want to perform [sc. the magic rite], take a little bit, light a coal, climb up on the roof, and burn the offering while saying this spell as [the moon] is rising, and immediately she⁴¹ will come.

λαβὼν μυγαλὸν ἐκθέωσον πηγαίῳ ὕδατι καὶ λαβὼν κανθάρους σεληνιακοὺς δύο ἐκθέωσον ὕδατι ποταμίῳ καὶ καρκίνον ποτάμιον καὶ στῆρ ποικίλης αἰγὸς παρθένου καὶ κυνοκεφάλου κόπρον, ἴβευς ὡὰ δύο, στύρακος δραχμὰς β', ζιμύρνης δραχμὰς β', κρόκου δραχμὰς β', κυπέρεως Ἰταλικῆς δραχμὰς δ', λιβάνου ἀτμήτου δραχμὰς δ', μονογενὲς κρόμμυον·

ταῦτα πάντα βάλε εἰς ὄλμιον σὺν τῷ μυγαλῷ καὶ τοῖς λοιποῖς καὶ κόψας καλλίστως ἔχε ἐπὶ τῶν χρειῶν ἀποθέμενος εἰς πυξίδα μολιβῆν. καὶ ὅταν βούλη πράττειν, ἀνελόμενος ὀλίγον καὶ ποιήσας ἀνθρακιᾶν ἀναβάς ἐπὶ δώματος ὑψηλοῦ ἐπίθουε λέγων τὸν λόγον τοῦτον ἀνατολῆς οὔσης, καὶ παραχρῆμα ἥξει.

GEMF 57 = PGM IV 2455–71 = 83 Betz, tr. O'Neil, modified

The recipe involves 'deifying' animals (a process that in practice means drowning them), grinding them up along with faeces and aromatic resins, and burning them on a coal to conjure Selene. As mentioned earlier, it is the element of the procedure attributed directly to Pachrates, and the kinds of materials it lists are also some of the most familiar from supernatural horror.

It is clear from the recipe that there are two broad classes of ingredients, vegetal and animal. Among the vegetal ingredients are pleasant-smelling materials like styrax, myrrh, saffron and frankincense, all widely attested in Egyptian temple worship, particularly in ointments for statuary, and along with saffron commonly used in the magical formularies for the consecration of ritual objects.⁴² The other ingredients are the more familiar, unpleasant kinds one might expect from supernatural horror. These are the dead animal parts, the insects, the bodily fluids and waste products. They resemble the *Dreckapotheke* or 'filth pharmacy' frequent in ancient Graeco-Roman pharmacology and medicine, including materials like blood, faeces, urine, semen and menstrual blood derived from humans and other larger animals, as well as various parts of smaller ones.⁴³ Sometimes, animals are 'made divine', i.e. drowned, for the purpose, while other times the procedure calls for animals or humans specified to have died in a certain way, usually violently. Such ingredients are common in recipes in the formularies. A Demotic recipe for a potion, for example, includes shavings from the head of a dead man, blood of a tick of a black dog, blood from your second and little finger, and your semen, along with some wine.⁴⁴ There are recipes with parts of cats, fingernails, donkey blood, milk in which falcons have been drowned, shell of crocodile eggs – it is a long list.⁴⁵ Occasionally these names may be code or *Decknamen* (code names) for other, often less disgusting materials.⁴⁶ The famous list of *hermēneumata* (ἐρμηνεύματα) or interpretations of names in P. Leiden I 384 are said to have been used to prevent the masses from knowing the true identity of the magical materials.⁴⁷ In this list, the filthy names are often code for plants or minerals, as crocodile dung is said to mean Ethiopian soil, or human bile is said to mean malt juice.⁴⁸ However, sometimes the materials are not so different from the code names. Lion's semen, for example, is interpreted as human semen, while blood of a baboon is said to be blood of a spotted gecko.⁴⁹

Pachrates' recipe is different from the parallel later in the text because it combines both kinds of ingredients, the pleasant and the unpleasant. In the parallel recipe, the materials are divided into beneficent (*agathopoion*, ἀγαθοποιόν) and coercive (*anankastikon*, ἀναγκαστικόν) offerings. The beneficent is to be burnt for two days and accompanied by a hymn to Selene. It includes the vegetal and aromatic incense compounds, which are mostly different from what we find in Pachrates: uncut frankincense, bay, myrtle, pyrēn, raisins, malabathron, costus.⁵⁰ These would be mostly pleasant-smelling. The coercive offering on the other hand consists of the animal materials, mostly the same in both versions, but which in the second version are to be combined and burned on the third day with a coercive spell, claiming that the victim has slandered the deity. Pachrates' recipe differs not only in the vegetal and aromatic ingredients, but in mixing all the ingredients together.

The encounter with Pachrates' recipe for the burnt offering is presented in the hymn. It is described primarily in terms of the smell of a desecrated offering to the goddess Selene.⁵¹ In fact, while the various *Dreckapotheke* of the magical formularies can have many magical (and allegorical, or even medical) meanings, the primary function of this procedure is to suggest that someone is offering material with the purpose of offending Selene. A third, coercive spell (*epanankos logos*, ἐπάνανκος λόγος) accompanying the

offering – used after the initial spell discussed earlier – offers a description of the scene that is embodied in the incense offering of Pachrates:

She, NN, is burning for you, goddess, some dreadful incense, and dappled goat's fat, blood and filth, the menstrual flow of virgin dead, heart of one untimely dead, the magical material of dead dog, woman's embryo, fine-ground wheat husks, sour refuse, salt, fat of dead dog, and mastic, and myrtle, dark bay, barley, and crab claws, sage, rose, fruit pits and a single onion, garlic, fig meal, a dog-faced baboon's dung, and egg of a young ibis. And this is sacrilege! She placed them on your altar; she set the flaming fire to juniper wood strips and slays a seahawk for you, a vulture, and a mouse, your greatest myst'ry, goddess.

ἡ δεινά σοι θύει, θεά, δεινόν τι θυμίασμα· αἰγός τε ποικίλης στέαρ καὶ αἷμα καὶ μύσαγμα, ἰχώρα παρθένου νεκρᾶς καὶ καρδίαν ἁώρου καὶ οὐσίαν νεκροῦ κυνός καὶ ἔμβρυον γυναικὸς καὶ λεπτὰ πίτυρα τῶν πυρῶν καὶ λύματα ὀξύεντα, ἄλα, στέαρ ἐλάφου νεκρᾶς σχῖνόν <τε> μυρσίνην τε, δάφνην ἄτεφρον, ἄλφιτα καὶ καρκίνιοιο χηλᾶς, σφάγνον, ρόδα, πυρῆνά τε καὶ κρόμμυον τὸ μόνον, σκόρδον τε, σύκων ἄλφιτον, κόπρον κυνοκεφάλιο ὥν τε ἴβεως νεᾶς – ἃ μὴ θέμις – τοῖς σοῖς ἔθηκε βωμοῖς, ξύλοις τε τοῖς ἀρκευθίνοις φλόγας πυρὸς βαλοῦσα ἰέρακα τὸν πελαγοδρόμον καὶ γῦπά σοι σφαγιάζει καὶ μυγαλόν, τὸ σόν, θεά, μυστήριον μέγιστον.

GEMF 57 = PGM IV 2574–92 = 86 Betz, tr. O'Neil

It is tempting to think that, in Pachrates' case, the defiled offering is mixed with the vegetal aromatic offering to make the desecration more convincing. Still, the reason some of the ingredients are offensive to the goddess, and hence the reason for the selection of ingredients, is not entirely clear. It is possible, as Ager has recently argued, that the unpleasantness of the scent may be behind the selection, the scent of materials like faeces being considered disgusting and offensive in contrast to the scent of aromatic offerings.⁵² There may, however, also be structural reasons and symbolic associations, like sympathies and antipathies, behind the selection. Jørgensen has pointed out parallels in early Egyptian mythological texts where baboons are offered to Selene.⁵³ Bortolani suggests what unites the fauna like scarabs (moon beetles), baboon, ibis and field mouse is that they are all sacred to different gods and so their unsanctioned sacrifice is sacrilegious.⁵⁴ Pachrates' mixture, therefore, is likely not only determined by some criterion of disgust relative to the practitioner or audience (moreover, they would likely find many scents disgusting, so there must be some further criteria that determine this particular mixture). It seems rather that the scents serve as evidence of a sacrilegious offering, meant particularly to offend Selene, and that the proper selection of these ingredients is what distinguished Pachrates' skill. At least, so the story goes, the demonstration worked well enough to convince Emperor Hadrian when he encountered it.

Magical vs medical horror

In the previous sections, we have seen how certain ingredients and procedures were associated with emotions of horror through the medium of pernicious unseen forces. Such forces, however, were not universally accepted. Particularly for those who believed the cosmos to be inherently good, like Platonists and some other philosophical schools, the horror if any of such acts and materials resulted from moral and aesthetic disgust. An especially clear example of such an attitude comes from Galen of Pergamum.

In one of Galen's work on pharmacy, *Simple Drugs*, he describes his medical predecessors' encounters with magical ingredients, code names and filth pharmacy. He discusses in particular the work of two compilers of medical lore, Pamphilus and Xenocrates of Aphrodisias, who he claims included magical materials and incantations in their pharmacological writings. Galen says Pamphilus collected information from Hermes the Egyptian about 'incantations, transformations and sacred herbs of decans and demons',⁵⁵ while Xenocrates, 'a man who did not live that long ago, but in our grandparents' generation, wrote about eating people when Roman Imperial law had forbidden it', as well as about faeces, menstrual blood, semen and other typically magical bodily materials.⁵⁶ He criticizes both for their superstition; however, while he finds fault with Pamphilus for writing about such materials without testing them, he criticizes Xenocrates for testing such disgusting things.

Galen is working in a broadly Platonist tradition of thinking about magic, where incantations and magical recipes are considered, as Fritz Graf puts it, 'psychological poisons'.^{*} Not only do they poison the minds of the victims with horror, but they also poison the minds of the practitioners by convincing them that they have control over the bodies of others and, even worse, over the gods.⁵⁷ Both victims and practitioners believe this, according to Plato, even though they have no evidence or understanding of how magic works and even though they know that engaging in such behaviours increases public distrust.⁵⁸ The problem, however, is pernicious, and Plato is resigned to the fact that people will believe in the power of demons; so rather than propose to educate it away, he thinks the state should implement laws to control the practice. Galen meanwhile is more determined to excise false beliefs, especially from medicine, which he normally does by presenting evidence that certain practices do not work. He does not think, however, that testing of the kind Hadrian engaged in is appropriate, at least for anyone who is not an emperor, since only an emperor in his view can justly condemn someone to die.⁵⁹

Galen explicitly claims not to have encountered or tested filth pharmacy himself in a medical context. This is because such materials are illegal, especially the use of human body parts in medicine, as well as the fact that they elicit disgust. We can understand his reaction as a kind of abject horror in the sense proposed by Julia Kristeva: the horror of blood and other bodily fluids arises when these materials go beyond the body's boundary and leave their proper place.⁶⁰ Galen writes, 'there are many parts and fluids in the bodies of animals that other people have written about, but that I myself have never experienced; for some of them are indecent and disgusting, some are also forbidden by law'.⁶¹ 'Other

people' refers to Xenocrates, a writer from Galen's grandfather's generation who Galen says writes as if he had tested 'which illnesses are treated by eating human brain, flesh or liver, which by drinking bones of the head, shin, or fingers, burnt or unburnt, and which by blood itself'; or as if he had tested 'drinking sweat, urine and menstrual fluid'; or earwax (he says, 'I have never been sick to the point where I would dare to ingest this'⁶²). Galen thinks all this material is 'indecent and disgusting' to varying degrees. He goes on to say, 'there is no one who would submit to an experience of these things while in their natural state, nor to what is more moderate than this (i.e. ingesting them) but still indecent, smearing some part of the body with faeces or human semen on account of some illness in it.'⁶³ The use of such materials occasions such feelings of abject horror and disgust in Galen that he thinks these materials should even be excluded from empirical study.

Therefore, Galen denies the forces that might lead to horrific dread of a Lovecraftian kind; nevertheless, he expresses abjection and moral disgust at the actions such beliefs lead to. The qualification of such acts as acts of 'horror', however, also depends for Galen on the social status of the one committing the act. He says that he came into possession of a book by Attalus III, the former king of Pergamum, who 'was extremely ambitious in acquiring experience of such things'.⁶⁴ Interestingly, Galen does not question the king's virtue, but claims if anyone else were to commit such acts, they would be horrific (*deinon*, δεινόν):

Concerning what are called *philtera* (love procedures), *agogima* (attraction procedures), *oneiropompa* (dream-senders) and *misetra* (hate procedures) – for I am deliberately using their words – I would not have mentioned them in writing to begin with, even if I had a great deal of experience with them, just as I would not mention deadly drugs or the ones they call *pathopoia* (disease-makers). For what is attributed to them is ridiculous: binding opponents so that they are unable to speak in court or causing a pregnant woman to miscarry or never to conceive or whatever else. In the majority of cases, it is impossible to prove that these things exist, while for some, even if it were possible, surely, they are harmful to people's well-being, so that I wonder how in heaven's name anyone ever came to the decision to write these things down. For how could they hope that the very acts that bring infamy to the living once they have been disclosed would bring them fame after their death? If, therefore, they had, as kings, conducted experiments on people who had been sentenced to death, they did nothing horrific (*deinon*). But since they came to write about these things as ordinary people without claim to such an office at any point in their lives, then either they have not tested anything and are writing about things they are ignorant of, or, if they have tested them, then they are the most impious of all people.

Τῶν δὲ καλουμένων φίλτρων, ἀγωγίμων, ὄνειροπομπῶν τε καὶ μισήτρων, αὐτοῖς γὰρ τοῖς ἐκείνων ὀνόμασιν ἐξεπίτηδες χρώμαι, τὴν ἀρχὴν ἂν, οὐδ' εἰ πείραν ἱκανὴν εἶχον, ἐμνημόνευσα διὰ γραμμάτων, ὥσπερ οὐδὲ τῶν θανασίμων φαρμάκων ἢ τῶν ὡς αὐτοῖ

καλοῦσιν παθοποιῶν. ἐκεῖνα μὲν γάρ αὐτῶν καὶ γελοῖα, καταδῆσαι τοὺς ἀντιδίκους, ὡς μηδὲν ἐπὶ τοῦ δικανικοῦ δυνηθῆναι φθέγξασθαι, ἢ ἐκτρώσαι ποιῆσαι τὴν κύουσαν, ἢ μηδέποτε συλλαβεῖν, ὅσα τ' ἄλλα τοιαῦτα. Τὰ μὲν γε πλεῖστα εἶναι τούτων ἐστὶ καὶ πρὸς τῆς πείρας ἀδύνατα ὑπάρχειν, ἔνια δὲ εἰ καὶ δυνατά, βλαβερὰ γοῦν γ' ἐστὶ τῷ βίῳ τῶν ἀνθρώπων, ὥστ' ἐγὼ νῆ τοὺς θεοὺς θαυμάζω κατὰ τίνα τὴν ἔννοιαν ἤκον ἐπὶ τὸ γράφειν αὐτὰ τινες. ἃ γὰρ καὶ τοῖς ζῶσιν ἀδοξίαν φέρει γνωσθέντα, πῶς ταῦτα μετὰ θάνατον εὐδοξίαν οἴσιν αὐτοῖς ἤλπισαν; εἰ μὲν οὖν βασιλεῖς ὄντες ἐν ἀνθρώποις ἐπὶ θανάτῳ κατακεκριμένοις ἐποιήσαντο τὴν τείραν αὐτῶν, οὐδὲν ἔπραξαν δεινόν. ἐπεὶ δ' ἰδιῶται τοιαύτης ἐξουσίας ἐν ὅλῳ τῷ βίῳ γεγονότες ἐπὶ τὸ γράφειν ἤκον αὐτὰ, δυοῖν θάτερον, ἢ μὴ πειραθέντες αὐτοὶ γράφουσιν ὑπὲρ ὧν οὐκ ἴσασιν, ἢ εἴπερ ἐπειράθησαν, ἀσεβέστατοι πάντων ἀνθρώπων εἰσιν.

Gal. SMT 10 proem, 12.251–253 K.

Galen here categorizes the effects recipes for horror are designed produce and he distinguishes two situations in which they might be attempted. The effects are similar to those we have seen. They include things like binding spells and sending dreams and Galen says he is using the names popularly given to them. The two situations he distinguishes both involve using these practices on living people. The difference, however, lies in who is conducting the experiments: a king or an ordinary person. In the case of the king, Galen sees ‘nothing horrific’ (*ouden deinon*, οὐδὲν δεινόν) in the act, so long as it is conducted on prisoners already sentenced to death. In the case of ordinary people, who have no claim on the lives of anyone else, Galen thinks their acts are not only wrong but extremely impious.

Galen’s claim that the king commits ‘nothing horrific’ (*ouden deinon*, οὐδὲν δεινόν) by testing procedures on prisoners sentenced to death gives some insight into what would make something a horror for Galen. It is not, as it is for Pachrates, powerful unseen forces that result in trembling and fear; rather what causes horror are the unknown intentions of ordinary people. Despite the fact that Galen, like Plato, is sceptical that magical practices can be rationally understood and that their unverifiability (or even unfalsifiability) discredits them as medical treatments, he nevertheless believes it is acceptable for those in absolute power to try to carry them out.⁶⁵ We can speculate, therefore, and wonder if Galen’s judgement that a particular act is horrific and impious is determined by a fear of those who wish to cause harm through hidden means who are not socially sanctioned to do so, even if nothing demonic is actually involved. What determines an act as horrific or not, therefore, would be closely connected to Galen’s ideas about legal and moral authority.

Encounter II: Replicating a blood and myrrh ink

The Recipe

Take blood of a night owl and myrrh ink, mix the two together, and with a new reed draw the figure, as appended, on a clean piece of paper; and having stared simultaneously at a clean wall, glancing to the east, after fastening it to a pure linen

cloth using thorns from a male date palm, veil the image completely. Having stepped back from it six cubits, after covering it, count to fifty-nine three times walking backwards, standing at the mark of the six cubits.

Λαβὼν αἷμα νυκτιβαούτος καὶ ζυρνομέλαν, ὁμοῦ τὰ δύο μίξας γράφε καινῶ καλάμῳ τὸ ζῳδιον, καθὼς περιέχει, εἰς πιττάκιον καθαρὸν, καὶ ἅμ' ἀτενίσας εἰς τοῖχον καθαρὸν, εἰς ἀνατολὴν βλέπων, πήξας εἰς σουδάριον ὀλόλινον σκόλοψιν ἄρρενικοῦ φοίνικος συνκάλυπτε τὸ ζῳδιον καὶ ἀποστὰς ἀπ' αὐτοῦ πήχεις ἕξ, συνκαλύψας μέτρησον πεντήκοντα ἑννέα ἐπὶ τρίς ἀναποδίζων, στήκων ἐπὶ τὸ σημίον τῶν ἕξ πηχῶν.

GEMF 68 = PGM XXXVI 264–74

We turn now to a modern encounter of a recipe for horror: an *epideixis* of a replication of a blood ink recipe. Galen may have found the idea of testing such recipes horrific or disgusting, but, in fact, replicating them can help us address questions about the use of horrible ingredients like blood in magic procedures more generally. As with Pachrates' recipe in the previous section, my choice here is somewhat arbitrary. Both recipes are similar insofar as they use bodily fluids and myrrh; both also suggest abject and supernatural horror. This recipe, however, is chosen mostly for its simplicity.⁶⁶ It involves only two ingredients, blood and myrrh ink. Moreover, it does not explain what the blood in the ink is for, which allows us to explore what role an ingredient like blood might have.⁶⁷ Some scholars have recently suggested that blood may have been used to improve an ink's writing properties.⁶⁸ I believe however that the recipe includes blood to add to its horror; for, when we actually follow the recipe and compare the resulting blood ink to a myrrh ink without blood, we find that the blood does not improve the ink as a writing medium. It does, however, seem plausible that it would contribute a sense of horror to its production and use. Even if we take claims about its magical efficacy asserted by the recipe's authors with a grain of salt, there is nevertheless something unsettling even today about blood as a constituent of ink, especially if it has no evident practical purpose.

Recreating the recipe

Replicating a recipe requires that we model its ingredients and methods closely, but a precisely detailed recreation is not always possible or expedient.⁶⁹ We also cannot always replicate using only one identification of any ingredient, since there may be different plausible interpretations of what the ancient terms refer to. In such cases, we need to test a variety of interpretations in order to understand how different materials behave. What follows is a detailed explanation of the procedure involved in replicating this recipe for horror. These details are necessary for ensuring the reproducibility of the experimental conditions that were chosen and for understanding how those conditions address the question of the purpose of blood in this recipe.

First, instead of using blood of a night owl, we can use blood that is more easily available, such as a drop of human blood.⁷⁰ This is because the chemically relevant part

of blood, its iron, does not differ much between human beings and other vertebrate animals.⁷¹ Moreover, variation in the kinds of blood used is not without precedent. One myrrh ink uses baboon blood (*haima kunokephalou*, αἷμα κυνοκεφάλου), which the interpretations of *GEMF* 15 = *PGM* XII discussed previously record as a code name for lizard blood.⁷² Another blood ink asks for blood of a human who died a violent death.⁷³ And in some cases, 'blood' is a code name for something other than blood, when it is used as a name for the red mineral *minium*, a common pigment for red inks.⁷⁴ Therefore, 'blood of a night owl' could already be the code name for some other animal's blood or for something that contains no blood at all. To simplify, we can blood that is easily available and set up a control that uses no blood at all.

Second, since the exact composition of the myrrh ink is unknown, it is necessary to determine a range of plausible mixtures to test. The myrrh in myrrh ink recipes can be myrrh that has been already burnt in a censer, or it can be unburnt.⁷⁵ If the myrrh in the ink is burnt myrrh, it would most likely be used to produce a carbon ink (discussed in what follows), but carbon inks require other ingredients to complete the ink, like binders and water. Likewise, if the myrrh is used raw or unburnt, again other ingredients would be needed to make a writable ink even if blood is added. Myrrh inks are common in magical Greek and Egyptian formulas and Christiansen and Blanco Cesteros assume that the myrrh in them is always burnt. Blanco Cesteros points to some additional passages in *GEMF* 57 = *PGM* IV where the ingredients of a magic ink, including myrrh, are first burnt as an offering.⁷⁶ I believe however we cannot rule out the possibility that the myrrh was supposed to be unburnt.⁷⁷ First, not all myrrh ink recipes claim to use burnt myrrh. Second, the scent of myrrh is important to its ritual use in other contexts, but the scent is removed if burnt. Third, there are myrrh ink recipes that have other sources of pigment, like iron-gall inks. So, to properly determine the role of blood in a myrrh ink, we need to use both myrrh ink from burnt and from unburnt myrrh.

As for other ingredients used in the myrrh ink, there are three possible mixtures corresponding to three kinds of ink that were in use at the time. First are carbon inks: early forms of black ink produced with water, pulverized carbonized organic matter, e.g. charcoal, such as from burnt myrrh, acting as a pigment, and something that binds the pigment to the writing surface, usually gum arabic, although blood has been reported to have been used for this purpose in other cultural contexts.⁷⁸ Most inks in fact that survive from antiquity are carbon inks. Second are iron-gall inks: produced from gallic and tannic acids, usually isolated from oak galls (Figure 12.1, left) by decoction, and iron salts, typically iron (ii) sulphate (Figure 12.1, right).⁷⁹ It is the presence of iron in these inks that connects them interestingly to the use of blood, which contains iron in relatively high amounts.⁸⁰ Third are mixed inks: carbon combined with iron-gall inks. The chemistry of iron-gall ink is important when thinking about whether myrrh is burnt or unburnt when it is used. If the ink includes burnt myrrh, then the myrrh ink the recipe refers to could either be a carbon ink or a carbon ink mixed with iron-gall. If the myrrh ink uses unburnt myrrh, however, then it would need to have contained some ingredient to make it legible, and iron-gall is a plausible candidate. The myrrh, then, would give the ink a distinct scent, rather than a dark pigment.



Figure 12.1 Constituents of iron-gall ink. Left: Oak galls from *Quercus infectoria* Oliv. 1801. Right: powdered iron (ii) sulphate or 'green vitriol'.

This makes for three kinds of myrrh ink (carbon ink with burnt myrrh, iron-gall ink with burnt myrrh, iron-gall ink with unburnt myrrh) that must be tested in comparison with their blood-containing counterparts. As a control, I also produced an iron-gall ink without myrrh and its blood-containing counterpart. There are, therefore, eight conditions to be tested:

1. CM: carbon ink (carbonized myrrh)
2. CM+B: carbon ink + blood (5:1)
3. G: iron-gall ink
4. G+CM: iron-gall ink + carbonized myrrh
5. G+UM: iron-gall ink + uncarbonized myrrh
6. G+B: iron-gall ink + blood
7. G+B+CM: iron-gall ink + blood + carbonized myrrh
8. G+B+UM: iron-gall ink + blood + uncarbonized myrrh

Materials and methods

Iron-gall blood ink test and preparation. Oak galls (from *Quercus infectoria* Oliv.) and iron (ii) sulphate were purchased in autumn 2021 from Kremer Pigmente in Germany.⁸¹

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One oak gall (~5g) was pounded using a mortar into pieces 2–3 millimetres in diameter. 5 grams ground oak gall was placed in a flask with 50ml boiling water and agitated intermittently for 10 minutes. 10 grams of iron (ii) sulphate was ground to a fine powder in a mortar and then 5 grams was placed in a flask with 50 millilitres boiling water and stirred until dissolved. Both solutions were allowed to cool. Several millilitres of venous blood were obtained through finger pricks using a sterile needle which were collected in a sterile vial. Several drops of water were added to prevent coagulation. Iron (ii) sulphate and blood solutions were added dropwise (total 0.5ml) to the gallic acid solution. No agitation was used.

Carbon ink and raw myrrh. Suhul myrrh, a spontaneous exudate from *Commiphora myrrha* (Nees) Engl., provenance Somalia, was purchased from a German resin supplier spring 2022.⁸² 10 grams were ground in a mortar to the consistency of wet sand, placed on aluminium foil, and suspended over a candle flame for one hour until a brittle, black, dry residue remained. 5 grams of the material was combined with 5 grams gum arabic and added to a mortar and ground to a fine powder. Water was added dropwise until a usable ink was produced. 5 grams of raw myrrh was added to another mortar with an equal amount of water and ground until a cream-coloured paste was produced.

Mixed blood inks. To produce the carbon and blood ink, 0.5 millilitres of blood was added to 2 millilitres of carbon ink. Iron-gall ink alone and iron-gall ink mixed with blood were each combined 1:1 with both carbonized and uncarbonized myrrh. Samples from each were dropped onto clean paper and then pulled along the surface using a clean dip pen and allowed to dry. Samples were photographed.

Results and discussion

An initial test was performed to determine whether human blood reacts with gallic acid to produce the dark-coloured pigment ferrous tannate (Figure 12.2, top). After several drops of iron (ii) sulphate solution were added to the gallic acid solution, there was an immediate reaction to a dark blue-black colour. When the same procedure was performed with several drops of blood, there was no reaction (Figure 12.2, centre). To make sure there was nothing wrong with the initial solution, several drops of iron (ii) sulphate were added to the blood and gallic acid mixture, which immediately produced a dark pigment (Figure 12.2, bottom).

The lack of any reaction when blood was added to the gallic solution suggests there was not enough available iron in the blood to produce an iron-gall ink. This is consistent with what one would expect based on vertebrate physiology. If there were free iron (ii) ions in the blood, it would be toxic to cells; for this reason, when iron is not bound to haemoglobin it is typically bound to other proteins like ferritin and stored in the liver.⁸³ There would not be freely available iron ions in the blood to produce the ink unless the iron-binding proteins themselves were denatured and destroyed.⁸⁴ Therefore it seems unlikely that blood was added to the myrrh ink in order to produce a darker pigment.⁸⁵

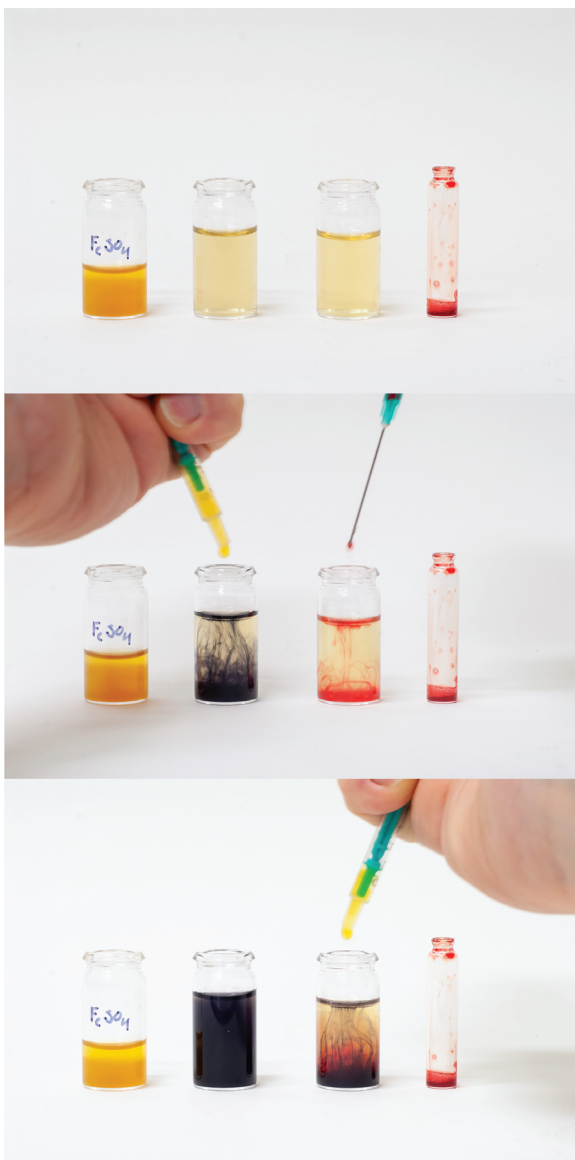


Figure 12.2 Experiment on the effects of blood and iron (ii) sulphate on a decoction of oak gall. Top photo shows (left to right) vial of iron (ii) sulphate solution (5g per 100ml), two vials oak gall decoction, and a small amount of the author's blood. Centre photo shows mixing of the reagents. On the left, iron (ii) sulphate reacts with gallic acids to form ferrous gallate, a blue-black pigment that reduces to insoluble ferric gallate in the presence of oxygen when dry. On the right, blood and the iron it contains does not react with the gallic acid to form any pigment. Bottom photo shows iron (ii) sulphate being added to the mixture of blood and oak gall decoction to show the solution could react. It seems unlikely blood was added to the mixture as a means of encouraging the formation of pigments.

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The uncarbonized myrrh when ground with water produced a cream-coloured, slightly pink paste. It initially dissolved easily, but a fraction (likely water-insoluble resin) required a longer period of grinding to go into suspension (Figure 12.3, left). The scent of the myrrh was strong, with characteristic balsamic, mushroom, astringent, and sweet-resinous notes. Carbonized myrrh mixed with gum arabic and water produced a jet-black suspension within a few seconds of grinding in the mortar (Figure 12.3, right). Its scent was earthy, somewhat smoky.

When this carbon ink was added to the blood, there was a visually striking mix of red and black. They did not immediately combine and it took agitation to incorporate them. In terms of its appearance on paper, the carbon and carbon-blood inks were visually indistinguishable. Both produced a rich, deep black ink (Figure 12.4a–b). They were clearly different, however, from the iron-gall inks, which were light to dark blue, at least while still wet (Figure 12.4c–h). There were also no visible differences between iron-gall inks made with blood and those made without.⁸⁶ The most significant visual difference, in fact, was not between blood and non-blood inks, but between mixed inks made with carbonized myrrh and the inks made with raw myrrh. The latter were noticeably lighter



Figure 12.3 Constituents of myrrh inks. Raw gum-resin from *Commiphora myrrha* (Nees) Engl. (top left) is ground with a small amount of water (bottom left). Burnt residue of *C. myrrha* (top right) is ground with a small amount of water and gum arabic as a binder (bottom right).

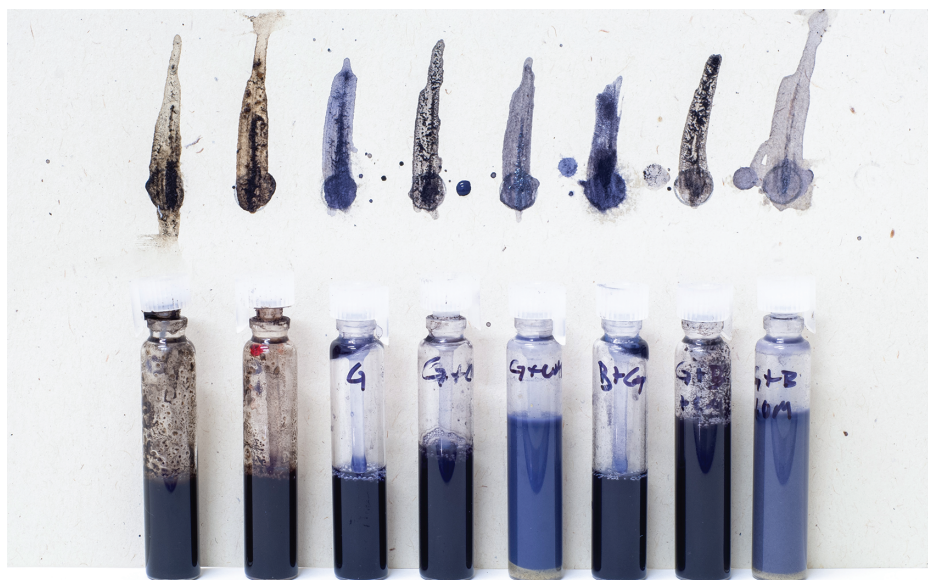


Figure 12.4 Myrrh and blood inks before agitation. From left to right: (a) burnt myrrh, gum arabic, water; (b) burnt myrrh, gum arabic, author's blood, water; (c) iron-gall ink; (d) iron-gall ink, burnt myrrh; (e) iron-gall ink, raw myrrh; (f) iron-gall ink, author's blood; (g) iron-gall ink, author's blood, burnt myrrh; (h) iron-gall ink, author's blood, raw myrrh.

in colour and strongly scented. In terms of ease of writing, none of the inks were well-made, nor did they write with any grace (due mostly likely to the skill of the experimenter). The scent of myrrh lingered on the paper in the case of inks produced with raw myrrh, while there was no noticeable difference in scent among the carbon, mixed or iron-gall ink without raw myrrh.⁸⁷

Experimental conclusions

From these experiments, it seems unlikely that the blood is added to the ink because of its contribution to it as an ink. In the carbon ink, iron-gall ink, and mixed ink, blood did not alter their writability. In fact, the experiments have shown that animal blood, at least in the amounts added here, is unlikely to be a practical source of iron for producing iron-gall ink, nor does blood noticeably darken either carbon ink or iron-gall inks at the concentrations tested. The superfluity of blood in an already constituted, useable ink, suggests it is added for a magical purpose whose testability lies outside the scope of these experiments.

Still, while its magical purpose cannot be tested, we can still consider the nature of the horror at play in this blood ink and in magical recipes more generally.⁸⁸ One way might be through Kristeva's notion of the abject: the disgust invoked by blood and other bodily

fluids when encountered outside their proper place. Kristeva's analysis of horror as abject begins from Old Testament discussions of ritual pollution caused by blood.⁸⁹ It is the fact that blood is polluting that allows her to claim that it is abject and then seek common structures that account for it. She argues, in essence, that a primordial fear of the loss of bodily integrity manifests as revulsion and leads to blood being considered polluting. While similar sentiments of revulsion are found in Galen, they are not obviously paralleled in the Greek and Egyptian magical formularies, which do not explicitly link blood and pollution. This is not because concepts of pollution are not at play in magical contexts. We have seen pollution used in the context of Pachrates' burnt offering, where it was connected to appropriate and inappropriate offerings to Selene. Purity conventions in ancient Egypt have what Joachim Quack calls more generally a 'delimiting function,'* determining where and when things, including bodily fluids, are allowed to be.⁹⁰ The myrrh ink recipe, however, does not make it clear where its authors or practitioners consider those boundaries to be; it may be that they extend beyond the boundaries of the victim's body. As Christiansen has noted, ingredients like blood are frequently used at this time and place: they could be a common product resulting from slaughter of sacred animals for the practice of offering animal mummies, a practice that took place on an industrial scale in Late Period Egypt.⁹¹ He also notes that blood is used in Egyptian medicine, which corresponds, as we have seen, with its use by Galen's medical contemporaries such as Xenocrates of Aphrodisias, who felt no horror using blood, semen, menstrual blood, or earwax in their treatments.

Another way we might understand the horror of blood in this recipe is through its sensorial aspect. When it was collected and before being incorporated into the ink, the blood was shiny, slimy and slippery, a deep red colour, smelling of metal. It did not incorporate easily into the ink, manifesting as a red frothy trace on the liquid's surface (cf. Figure 12.4b). Its shining colour and thickness, especially when used in large quantities, is unsettling in a way that is difficult to express. The experience is similar to the effect of sensorial horror in contemporary film, where close-ups of blood and emphasis on its tactile qualities can be used to elicit revulsion, disgust and titillation.⁹² However, there is little textual evidence in the Greek and Egyptian magical formularies for sensorial horror of blood. Blood itself is often mentioned, but almost always as an ingredient without comment, no more unsettling than any other. This may be because, as Blanco Cesteros has suggested, blood is often a code name for other, less horrifying, materials.⁹³ Yet, at the same time, code names referring to blood might also suggest that the sensorial (or even abject) sense of horror elicited by blood are precisely what led to it being used as a code name in the first place: to keep the uninitiated away from performing the recipes due to the horrific nature of the ingredients.

One final way of understanding the horror of the blood ink is supernatural. This is the kind of horror that seems most absent from the performative context of a scientific experiment as presented in this chapter. Blood used and described as an experimental condition is no more or less demonic than any other material. The use of blood for ritual purposes, however, is a cause and source of horror, in a Lovecraftian sense, in the magical

recipe for a myrrh blood ink; and the same holds for other recipes involving distorted and dismembered bodies. Their parts are required by the supernatural entities that the practitioners are trying to manipulate and control, while their victims are simply instruments used to fulfil those aims. The absence of supernatural horror in a replication, however, is not only a difference between a modern experience of these recipes and an ancient one. For Galen, too, the items of the *Dreckapotheke* were objects of abject horror without any hint that he considered they might be able to suspend the fixed laws of nature.

Conclusion: Encountering recipes for horror

This chapter began with several tensions: between supernatural and bodily horror and between textual transmission and cultural continuity. As we have seen, among Greek and Egyptian magical sources, what Lovecraft called ‘the dread of outer, unknown forces’ is often localized and centred in the bodies of its victims. This is evident both from descriptions of the kinds of fear that demonic magic is meant to produce and in the bodily distortions and mutilations their procedures describe. The targets of the procedures, the *Dreckapotheke* and other disgusting ingredients they use from animal and human bodies, and occasionally even the practitioners themselves, become designated places where such forces are manifested.

The body in fact is a critical site where our sources suggest unseen forces are experienced and evaluated – either as reliable and governable, as seen with Pachrates and Hadrian, or as existing things in the world, as seen with Galen. Moreover, the focus on display or *epideixis* involving the body and its manipulation suggests that the fear of these forces goes beyond a fear of spiritual or unseen actors outside of our control. The world of fixed laws and the world of demons is perhaps not as distant as Lovecraft suggests. The existence of magical formulae and a common set of magical materials implies that these procedures were taken to be reliable in much the same way as procedures that make use of everyday forces of nature. This holds even though their reliability was imperfect and even though writers like Plato or Galen doubted the possibility of manipulating the unseen world. In either case, violence was inflicted on victims in the service of confirming or disconfirming such claims.

Even in the sterilized context of a laboratory replication, where replicability is of central importance, there is a dread – albeit muted – upon discovering that blood may have served no practical purpose as a material in itself but was instead used to call on malign forces. Blood may be abject, evoking disgust through a recognition of our own bodily distortion. However, in this context, it also brings an expectation of something fearful through its presence. The expectation seems not to be due to any innate sense of what blood is but rather to the continuation of an ancient tradition of mediating desire through bodily violence. Thus, the horror found in these ancient recipes is not just about uncontrollable agents but also about the tangible, violent experiences the recipes evoke and the deep cultural roots they dimly trace.

Notes

1. This chapter was written as part of the grant project GM21-30494M ‘Alchemies of Scent’ supported by the Czech Science Foundation and coordinated by the Institute of Philosophy of the Czech Academy of Sciences in Prague. Thanks to participants of the conference, ‘Medical Knowledge and its “Sitz im Leben”: Body and Horror in Antiquity’, for helpful discussion, especially Nick Lowe and Glenn Most; special thanks to Dana Zentgraf for acquiring materials for a practical demonstration of the ink on this occasion. I am grateful to Victor Golubev and Laura Prieto for help with the experiments, and to Miriam Blanco Cesteros, Thomas Christiansen, George Kazantzidis, Diana Mičková and Maddalena Rumor for comments on various drafts. I am especially grateful to Ronja Hildebrandt and Chiara Thumiger for discussion and extensive comments on the final draft. Errors are my own.
2. Arist. *De an.* 1.5, 411a8. He attributes it to Thales, but at Arist. *Gen. an.* 3.11, 762a21, he agrees at least in spirit when he says, ‘in a way all things are full of soul’. I do not mean to suggest ancient Graeco-Roman thinkers denied there was regularity in the cosmos (this is a fact to be explained, not something that can reasonably be doubted); rather, my point is that they did not use the idea of fixed laws to explain it. This strategy of explanation does not appear in Europe until the sixteenth and seventeenth centuries. On this see Menn 2000: 139–43. For Lovecraft on classical sources, see Moreland 2018. Throughout this chapter, I follow the abbreviations of the Oxford Classical Dictionary, fourth edition.
3. This serves as my working definition of supernatural horror. It is however only one way of thinking about horror. I discuss Kristeva’s characterization of horror as ‘abject’ below. As Thomas Christiansen points out to me, while horror involves disgust, it also involves fascination and attraction, of which the recipes in this chapter (and the chapter and volume itself) are examples. I can only touch on these themes here.
4. As Miriam Blanco Cesteros suggests to me, some lines of transmission likely begin from Hellenistic Greek and especially post-Augustan Latin descriptions of witches, e.g. Hor. *Ep.* 5 and Apul. *Met.* 3.17. These lines are explored in Ager 2022.
5. Many of these texts are included in a new collection, the *Greek and Egyptian Magical Formularies* (*GEMF*), edited by Christopher Faraone and Sofia Torallas Tovar (Faraone and Torallas Tovar 2022, hereafter FTT). I have used these editions when published (only one volume so far of several proposed). For the rest, I have made use of the collection of *Greek Magical Papyri* (*Papyri Graecae Magicae* or *PGM*), edited and translated into German under Karl Preisendanz’ editorship in the mid-twentieth century (Preisendanz 1928, 1931) and translated into English by numerous scholars under Hans Dieter Betz’s editorship at the University of Chicago a few generations later (Betz 1986). The magical papyri are often written in multiple languages, including Greek, Egyptian (in hieratic, Demotic and Coptic scripts) and cipher. The *GEMF* volumes edit all languages, while the *PGM* printed only the Greek language portions. This was partly corrected by Betz’ inclusion of translations of the *Magical Demotic Papyri* (*PDM*). Recipes will be cited by *GEMF* or *PGM/PDM* number and line number, followed by page and line number of the edition. The reader should keep in mind that despite being included in collections, the nature of the texts themselves and their relation to one another is still unclear and a matter of ongoing scholarly investigation. Other texts, mostly from medical authors, will be introduced along the way using standard conventions.
6. I am using the term ‘recipe’ to describe any set of instructions for combining ingredients as part of magical or ritualistic procedures, including rites, rituals, incantations and prayers. These are essentially texts that guide someone in performing a specific action, falling into the category of what might be termed *Gebrauchstexte* (utilitarian texts). Despite occasionally

- featuring unusual or highly technical language, these recipes (or procedures, as I will use interchangeably) offer a pragmatic window into the practices they describe. On recipes in general, see Lehmhaus and Martelli 2017. On recipes as fluid texts, see Raggetti 2018.
7. There are obviously many literary sources that depict encounters with magic, even magic recipes, but I limit my focus to pragmatic texts themselves (although the boundary can at times be fuzzy). It seems to me that magical recipes and medical texts occupy a distinctive place in the corpus of ancient texts, since they record the practical aspect of how people in the ancient Mediterranean world sought to understand, harness and protect themselves from unseen forces.
 8. For recent examples, see Raggetti 2021; Marchini et al. 2022.
 9. Tullett 2023: 31–5.
 10. The papyrus been extensively studied by experts of ancient magic, most recently by Dieleman and Sarischouli. Agathokles' dream-sender is one of twenty-nine recipes written on the back (*verso*) of a scroll whose front (*recto*) preserves an ancient Egyptian novel called *The Myth of the Sun's Eye*. On palaeographical grounds, Dieleman and Sarischouli, who published an edition with others as *Greek and Egyptian Magical Formulary 15 (GEMF 15)*, see Dieleman, Sarischouli et al. 2022), suggest it dates from the second century CE, while earlier scholars had dated it to the fourth century (Dieleman and Sarischouli 2022: 67). They also think that the collection is the work of several scribes, who wrote a corpus using a mixture of languages and scripts, primarily Demotic and Greek, but also hieratic and cipher (Dieleman and Sarischouli 2022: 64). The papyrus came to the attention of European collectors when, in 1828, it was purchased on behalf of the Dutch government as part of a set of Egyptian relics. It was catalogued as Leiden papyrus I 384, and later edited and published (the Greek text anyway) as part of Preisendanz' *Papyri Graecae Magicae* XII.
 11. 'NN' is used in translation of Graeco-Egyptian magic as a generic proper-name placeholder for a spell's target. It may derive from the Latin phrase *nomen nescio* ('I do not know the name'), *nomen nominandum* ('name to be named'), or even *nomen notetur* ('name to be noted'). It is a modern convention, translating the ancient Greek word 'δεῖνα', which just means 'so-and-so'. For the history of the convention, see Dosoo and Preininger 2023: 33.
 12. Ciruolo 2001: 279.
 13. These terms were explored by Lowe 2021.
 14. On *phrikē* (φρίκη), see Cairnes 2015.
 15. *Iunx* (ἰϋγξ) referred first to the wryneck, a kind of bird that as part of a magical procedure was tied to a revolving wheel which somehow attracted the victim; later it came to refer to the wheel itself spun with a string but without the bird, and finally to any attracting charm or spell (Faraone 1999: 25). It features prominently in Theocritus' *Idyll* 2 called the *Φαρμακεύτριαι* or 'The Sorceresses'. While it does not appear in *GEMF* or *PGM* collections, *iugges* are associated with Cleopatra VII, queen of Egypt, who ruled 51–30 BCE (e.g. Suda ι 761 and see Faraone 1999: 121).
 16. The text of the procedure is found in the Paris Magical Codex, Bibliothèque nationale de France papyrus Supplément grec 574, which Preisendanz included in his collection as *PGM* IV and is numbered as *GEMF* 57 by Faraone and Torallas Tovar. Scholars of magical texts suggest it is a fourth-century CE codex, and note that it is one of the longest ancient Graeco-Egyptian magical texts found, at 72 pages in length (Love 2016: 4, LiDonnici 2022: 170). Although there is some debate, Korshi Dosoo and Edward Love have argued it is part of the Theban Magical Library, a collection of magical texts in Greek and Demotic found in the nineteenth century and part of the set of Egyptian relics acquired by European collectors starting in 1828 and 1857 that I mentioned earlier (Love 2016: 8, Dosoo 2016: 259). The

codex itself has been described as a ‘library’ by Lynn LiDonnici in her study of its structure, because the compiler seems to have collected and categorized texts according to internal similarities, such as the divinity named in the procedure, etc. (LiDonnici 2022: 169).

17. The deity is called Aktiōphis (Ἀκτιῶφης, *GEMF* 57 = *PGM* IV 2473), an epithet of Selene (*GEMF* 57 = *PGM* IV 2525, 2558, 2569), but she is addressed by other names as well, like Artemis (*GEMF* 57 = *PGM* IV 2523), Persephone (*GEMF* 57 = *PGM* IV 2524), Aphrodite (*GEMF* 57 = *PGM* IV 2557) and the Mesopotamian goddess Ereškigal (Ερεσχιγαλ, *GEMF* 57 = *PGM* IV 2484; see Schwemer 2018: 66). LiDonnici argues (convincingly I think) that the spells (*logoi*) or hymns likely predate the procedures and show a syncretism of female, lunar deities including Selene, Artemis, Hecate and Persephone along with Aphrodite (LiDonnici 2022: 188–90). As Chiara Thumiger points out to me, Selene and Hecate also figure prominently in Theocritus’ *Idyll* 2 and the rites there likewise involve burnt offerings (although with different ingredients). On this, see Faraone 1999: 50.
18. *hē deīna*, ἡ δεῖνα, e.g. at *GEMF* 57 = *PGM* IV 2476, 2478, 2484.
19. Escolano-Poveda 2020: 132.
20. See Erman 1890: 10–11. The text has been recently translated by Diana Míčková and Dorotea Wollnerová (2019a: 77–111), with interlineal glosses available at Míčková and Wollnerová 2019b. Similar examples are found in the Demotic story of Setne II (Lichtheim 1980: 142–4) and in *Exodus*’ story of Aaron and pharaoh’s magicians turning sticks into snakes at *Exod.* 7:8–13. Thanks to Diana Míčková for discussion on these points.
21. On the term, see Escolano-Poveda 2020: 90.
22. The parallel attraction procedure and incense offering in the next entry at *GEMF* 57 = *PGM* IV 2674–94 contains some but not all the same ingredients. This suggests to me that the selection of materials is a matter of expertise.
23. Edmonds 2020: 35.
24. Totelin 2004, Mayor 2010.
25. See Bubb 2022: 58–76, Salas 2021: 29–30, von Staden 1997: 37–9.
26. von Staden 1997: 39.
27. *Gal. Lib. Prop.* 3.15–18, 19.22 K. = 144,19–145,15 Boudon-Millot.
28. Salas 2021: 37–40, Bubb 2022: 70. Some of these have now been re-enacted by the Atlomy Project, funded by the European Research Council and headed by Orly Lewis at the Hebrew University of Jerusalem. See Pelavski et al. 2024.
29. For detailed discussion, see Escolano-Poveda 2020: 181–4.
30. Lucian, *Philops.* 34.
31. Ogden 2004: 101–10.
32. Lucian *Philops.* 34–6.
33. *GEMF* 16.675–82 = *PDM* XIV 675–94 [*PGM* XIVc 16–27] = FTT 222,1–8.
34. Pachoumi 2017: 134.
35. Faraone 1999; see also Edmonds 2019.
36. Faraone 1999: 50.
37. Faraone 1999: 60.
38. Faraone 1999: 59.
39. Pachoumi 2017: 7, Quack 2018, Nagel 2018.
40. Nagel 2018: 144, Quack 2018: 204.

41. Here, most likely the goddess rather than the victim, since the spell (quoted previously) begins by asking Aktiōphis/Selene to appear.
42. One ingredient, Italian cyperus (*cyperus Italikos*, κύπερος Ἰταλικός), is otherwise unattested; no other form of cyperus appears in the magical papyri. For ointments in statuary, see Wilde et al. 2025.
43. Rumor 2015; 2020.
44. *GEMF* 16 = *PDM* XIV 428–35 = 192,5–8 FTT
45. For flacons, see *GEMF* 31 = *PGM* I 1–5; for crocodile eggs, *GEMF* 16 77–8.
46. See e.g. Bortolani 2016: 304, Rumor 2020, Blanco Cesteros 2020.
47. See, however, Blanco Cesteros 2020, who argues that the names are not coded in order to hide them, but technical Greek jargon of which there are traces in Dioscorides.
48. *GEMF* 15 462, 472.
49. *GEMF* 15 463
50. *GEMF* 57 = *PGM* IV 2670–90. See Ager 2022: 82–3.
51. On these scents and how they were received, see Ager 2022: 79–85.
52. Ager 2022: 82–3.
53. Jørgensen 2015: 149.
54. Bortolani 2016: 304–5.
55. Gal. *SMT* 6 pr., 11.797–8 K.: *epooidas kai metamorphōseis kai dekanōn kai daimonōn hieras botanas*, ἐπωιδὰς καὶ μεταμορφώσεις καὶ δεκανῶν καὶ δαιμόνων ἱερὰς βοτάνας.
56. Gal. *SMT* 10 pr., 12.284 K.: *ho Xenokratēs, anthrōpos ou palai gegonōs, alla kata tous pappous hēmōn, tēs Rōmaikēs basileias apēgoreukuias anthrōpous esthiein*, ὁ Ξενοκράτης, ἄνθρωπος οὐ πάλαι γεγονῶς, ἀλλὰ κατὰ τοὺς πάππους ἡμῶν, τῆς Ῥωμαϊκῆς βασιλείας ἀπηγορευκυίας ἀνθρώπους ἐσθίειν. On Galen's relation to Pamphilus and Xenocrates, see Petit 2017. On the context of Xenocrates' disgusting materials, see Petit 2020.
57. Graf 2002: 97–9.
58. Pl. *Leg.* 11, 932E–933B.
59. Gal. *SMT* 10 pr., 12.252 K.
60. Kristeva [1982] 2024.
61. Gal. *SMT* 10 pr., 12.248 K.
62. Gal. *SMT* 10 pr., 12.248–9 K.
63. Gal. *SMT* 10 pr., 12.249 K.
64. Gal. *SMT* 10 pr., 12.251–2 K.
65. It should be noted, as Caroline Petit has shown, that Galen sometimes does admit the effectiveness of cures like amulets in *SMT*. See Petit 2017: 63–6.
66. Thanks to Thomas Christiansen for pressing me to clarify the choice of this recipe.
67. On blood in ancient magic, see Watson 2019: 138.
68. Miriam Blanco Cesteros 2021: 39 points out that the use of blood is not restricted to ritual or magical contexts and that it can be a source of red pigment (cf. Scott et al. 1996, Yu 2020). Research by Bicchieri and Pinzari 2016 has found traces of erythrocytes (red blood cells) on the surface of a tenth-century non-magical manuscript from Yemen that was written using iron-gall inks and hypothesize that the blood acted as an unusual source of iron for making the ink. To test their hypothesis, they produced iron-gall inks using Aleppo oak galls mixed

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with either iron (ii) sulphate or a combination of iron (ii) sulphate and human blood and found all three produced an ink, with the blood inks resulting in darker blacks.

69. Fors et al. 2016.
70. Many myrrh and blood inks source their blood from birds, e.g. white dove (*peristeras leukēs haima*, περιστερᾶς λευκῆς αἷμα, *GEMF* 57 = *PGM* IV 2888; *GEMF* 72 = *PGM* VIII 69), crow (*haima korōnēs*, αἷμα κορώνης, *GEMF* 72 = *PGM* VIII 69), quail (*haima ortugiou*, αἷμα ὀρτυγίου, *GEMF* 15 = *PGM* XII 145).
71. Doty and Lednev 2018.
72. For baboon blood recipe, see *GEMF* 60 = *PGM* XIII 316.
73. *GEMF* 57 = *PGM* IV 2201.
74. Blanco Cesteros 2021: 50–5. Many thanks to Thomas Christiansen for bringing this possibility to my attention. In the replication that follows, I do not test red mineral pigments, but it is nevertheless a possible interpretation worth an experiment.
75. Myrrh is a common writing medium in the Greek and Egyptian magical formularies, probably because of its magical and ritualistic symbolism. Myrrh is associated, for example, with Selene at *GEMF* 60 = *PGM* XIII 20.
76. Blanco Cesteros 2021: 37–8.
77. Blanco Cesteros (forthcoming) has independently carried out similar experiments with raw myrrh as part of her project, *INKQuery*, focusing especially on the taste of the ink, which is excessively bitter.
78. Scott et al. 1996. Blanco Cesteros 2021: 38 suggests a binder may have been intentionally lacking in many magical inks; she points out this may be because they were meant to be temporary.
79. Iron (ii) sulphate is also called ‘green vitriol’. The word in ancient Greek is ‘*chalchanthos*’ (χάλκανθος). A myrrh ink recipe is preserved in the *Greek and Egyptian Magical Formularies* and is useful for reconstructing the procedure. The ink is used in a charisma procedure. It contains 1 part myrrh, 4 parts misy (mistranslated as ‘truffle’ in Betz, perhaps iron sulphide, see Christiansen 2017: 184), 2 parts *chalchanthos*, an impure source of iron (ii) sulphate, 2 parts oak gall, and 3 parts gum arabic (*GEMF* 15 448 = *PGM* XII 400 = 118,1 FTT).
80. The inks are produced when iron ions bind to the organic gallic and tannic acids to form complexes of ferrous gallate or tannate, which, while drying, react with atmospheric oxygen to produce ferric gallate and tannate, which are dark in colour and insoluble in water; see Díaz Hidalgo et al. 2018.
81. Kremer Pigmente GmbH & Co. KG (Hauptstr. 41–7, 88317 Aichstetten DE, www.kremer-pigmente.com).
82. Georg Huber of Mothers Goods (Kalterer Str. 29, 64646 Heppenheim DE, mothersgoods.com).
83. Muñoz et al. 2011.
84. One could do this with heat, but heat it is not indicated in the recipe.
85. No spectroscopic analysis on the inks was performed and thus results from the two experiments are not directly comparable. However, the comparison of one drop iron (ii) sulphate to one drop blood suggests that blood is not a suitable source of iron for producing the inks. In their procedure, Bicchieri and Pinzari 2016: 572 combined a standard iron-gall ink with blood in a 2:1 ratio. But in this case, even if there were free iron in the blood, it is unclear how much unreacted gallic or tannic acids would have been present when the blood

was added, but we would assume there would need to be unreacted gallic or tannic acids to make the ink darker.

86. This may be due to the use of lower concentrations of blood than in the experiments by Bicchieri and Pinzari.
87. No spectroscopic measurements were made of the inks in liquid form or dry on the paper. While I think the value of such measurements for assessing visual variations is dubious, I admit that, if some variation is shown, the process may be replicable and made more efficient.
88. Kieschnick 2000 explores a similar question, although not focused on horror, but the ascetic aspects of blood writing found among Chinese Buddhist scribes. There, according to Kieschnick, and similar to what we see in *GEMF*, blood is a flexible material: a source of uneasiness, fear, morbid fascination as well as ritually and spiritually useful (Kieschnick 2000: 190). Thanks to Thomas Christiansen for pointing this out to me.
89. Kristeva [1982] 2024: 97–100.
90. Quack 2013: 152.
91. Christiansen 2017: 180–1.
92. Powell 2012: 172–3.
93. Blanco Cesteros 2021: 51–3.

